Homework 04: Solutions

REAL VARIABLES I

Fall 2026

Solutions for Homework 04.

1. (Completeness)

1.a

Remark (Solution). Definition (1) says: every subset of every $N \in \mathcal{E}$ with $\mu(N) = 0$ belongs to $\mathcal{E}$. Definition (2) says: if $A \in \mathcal{E}$, $B \subseteq X$ and $A \triangle B \subseteq N$ for some $N \in \mathcal{E}$ with $\mu(N) = 0$, then $B \in \mathcal{E}$. We show (1)⇒(2) and (2)⇒(1).
(1)⇒(2). Let $A \in \mathcal{E}$, $B \subseteq X$, and $A \triangle B \subseteq N$ with $N \in \mathcal{E}$, $\mu(N) = 0$. The sets $A \setminus B$ and $B \setminus A$ are subsets of $A \triangle B$, hence of N. Therefore, according to (1), it holds that $A \setminus B \in \mathcal{E}$ and $B \setminus A \in \mathcal{E}$. Since $\mathcal{E}$ is a σ-algebra, $A \setminus (A \setminus B) = A \cap B \in \mathcal{E}$. Hence $B = (A \cap B) \cup (B \setminus A) \in \mathcal{E}$, as a union of two sets in $\mathcal{E}$.
(2)⇒(1). Let $N \in \mathcal{E}$ with $\mu(N) = 0$ and let $B \subseteq N$. Take $A = \emptyset$. Then $A \in \mathcal{E}$ and $A \triangle B = B \subseteq N$. Therefore, according to (2) applied to A, B and N, it holds that $B \in \mathcal{E}$.

1.b

Remark (Solution). By Carathéodory's theorem, $\mathcal{M}_{\mu^*}$ is a σ-algebra and $\mu = \mu^*|_{\mathcal{M}_{\mu^*}}$ is a measure on it. Hence it remains to show definition (1): if $N \in \mathcal{M}_{\mu^*}$, $\mu(N) = \mu^*(N) = 0$ and $B \subseteq N$, then $B \in \mathcal{M}_{\mu^*}$. By the definition of Carathéodory measurability, $B \in \mathcal{M}_{\mu^*}$ means $\mu^*(T) = \mu^*(T \cap B) + \mu^*(T \setminus B)$ for every $T \subseteq X$.
Lemma. If $Z \subseteq X$ and $\mu^*(Z) = 0$, then $\mu^*(T) = \mu^*(T \cap Z) + \mu^*(T \setminus Z)$ for every $T \subseteq X$.
Proof. Let $T \subseteq X$. Since $T \cap Z \subseteq Z$, monotonicity gives $\mu^*(T \cap Z) \leq \mu^*(Z) = 0$. Since $T \setminus Z \subseteq T$, monotonicity gives $\mu^*(T \setminus Z) \leq \mu^*(T)$. Hence

$$\mu^*(T \cap Z) + \mu^*(T \setminus Z) \leq 0 + \mu^*(T) = \mu^*(T).$$

The reverse inequality $\mu^*(T) \leq \mu^*(T \cap Z) + \mu^*(T \setminus Z)$ is subadditivity, since $T = (T \cap Z) \cup (T \setminus Z)$.
Now $B \subseteq N$, so monotonicity gives $\mu^*(B) \leq \mu^*(N) = 0$. Therefore, according to the Lemma with $Z = B$, it holds that $B \in \mathcal{M}_{\mu^*}$.

2. (Distribution functions and probability measures)

2.a

Remark (Solution). Let $\mu \in \mathcal{P}([0,1))$ and $F = F_\mu$. We show that F is nondecreasing with $F(0) = 0$ and $F(1) = 1$, that $F(t) = \lim_{s \uparrow t} F(s)$ for every $0 < t \leq 1$ (so $F \in \mathcal{D}$), and that, for $t \in [0,1)$, F is discontinuous at t if and only if $\mu(\{t\}) > 0$.
F is nondecreasing with $F(0) = 0$, $F(1) = 1$. $F(0) = \mu([0,0)) = \mu(\emptyset) = 0$ and $F(1) = 1$ by definition. Let $0 \leq s \leq t < 1$. Then $[0,s) \subseteq [0,t)$, so monotonicity of μ gives $F(s) \leq F(t)$. Also $F(s) \geq 0$ and $F(t) \leq \mu([0,1)) = 1 = F(1)$. Hence $0 \leq F(s) \leq F(t) \leq F(1) = 1$, and F is nondecreasing with values in $[0,1]$.
Left-continuity. Let $0 < t \leq 1$. Since F is nondecreasing, $\lim_{s \uparrow t} F(s)$ exists and equals $\lim_n F(s_n)$

for every nondecreasing sequence $s_n < t$ with $s_n \to t$. For such a sequence, $[0, s_n) \subseteq [0, s_{n+1})$ and $\bigcup_n [0, s_n) = [0, t)$. Hence, by continuity from below,

$$\lim_{s \uparrow t} F(s) = \lim_n \mu([0, s_n)) = \mu([0, t)).$$

If $t < 1$, then $\mu([0, t)) = F(t)$ by definition. If $t = 1$, then $\mu([0, 1)) = 1 = F(1)$. In both cases $\lim_{s \uparrow t} F(s) = F(t)$. Thus $F \in \mathcal{D}$.

Right limit. Let $t \in [0, 1)$. We show that $\lim_{s \downarrow t} F(s) = F(t) + \mu(\{t\})$. Since F is nondecreasing, the limit exists and equals $\lim_n F(s_n)$ for every nonincreasing sequence $s_n \in (t, 1)$ with $s_n \to t$. For such a sequence, $[0, s_{n+1}) \subseteq [0, s_n)$ and $\bigcap_n [0, s_n) = [0, t]$. Moreover $\mu([0, s_0)) \leq 1 < \infty$. Hence, by continuity from above,

$$\lim_{s \downarrow t} F(s) = \mu([0, t]) = \mu([0, t)) + \mu(\{t\}) = F(t) + \mu(\{t\}).$$

Discontinuity criterion. Let $t \in [0, 1)$. If $t \in (0, 1)$, then F is left-continuous at t by the previous part. Hence F is continuous at t if and only if $\lim_{s \downarrow t} F(s) = F(t)$. If $t = 0$, then continuity of F on $[0, 1]$ at 0 means exactly $\lim_{s \downarrow 0} F(s) = F(0)$. In both cases, by the formula for the right limit, F is continuous at t if and only if $F(t) + \mu(\{t\}) = F(t)$. This holds if and only if $\mu(\{t\}) = 0$, since $F(t)$ is finite.

2.b

Remark (Solution). Let $\mathcal{P} \coloneqq \{[0, t) : 0 \leq t \leq 1\}$. We show that $\mathcal{P}$ is a π-system with $[0, 1) \in \mathcal{P}$ and $\sigma(\mathcal{P}) = \mathcal{B}([0, 1))$. The coincidence criterion for finite measures states: if $\mathcal{P}$ is a π-system, μ, ν are finite measures on $\sigma(\mathcal{P})$, $\mu = \nu$ on $\mathcal{P}$, and $\mu(X) = \nu(X)$, then $\mu = \nu$ on $\sigma(\mathcal{P})$. Here $X = [0, 1)$.

Lemma. $\mathcal{P}$ is a π-system and $\sigma(\mathcal{P}) = \mathcal{B}([0, 1))$.

Proof. For $s, t \in [0, 1]$, $[0, s) \cap [0, t) = [0, \min\{s, t\}) \in \mathcal{P}$, so $\mathcal{P}$ is a π-system. Each $[0, t) = (-\infty, t) \cap [0, 1)$ is open in $[0, 1)$. Hence $\mathcal{P} \subseteq \mathcal{B}([0, 1))$, and so $\sigma(\mathcal{P}) \subseteq \mathcal{B}([0, 1))$.

For the converse inclusion, let $U \subseteq [0, 1)$ be open in $[0, 1)$. For each $x \in U$ there are rationals $a < x < b$ with $(a, b) \cap [0, 1) \subseteq U$. Hence U is the union of those sets $(a, b) \cap [0, 1)$ with $a < b$ rational and $(a, b) \cap [0, 1) \subseteq U$. This union is countable. Since $\sigma(\mathcal{P})$ is closed under countable unions, it suffices to show that $(a, b) \cap [0, 1) \in \sigma(\mathcal{P})$ for all rationals $a < b$. We distinguish three cases.

- If $b \leq 0$ or $a \geq 1$, then $(a, b) \cap [0, 1) = \emptyset \in \sigma(\mathcal{P})$.
- If $a < 0 < b$, then $(a, b) \cap [0, 1) = [0, \min\{b, 1\}) \in \mathcal{P}$.
- If $0 \leq a < 1$ and $b > a$, then

$$(a, b) \cap [0, 1) = [0, \min\{b, 1\}) \setminus [0, a] = [0, \min\{b, 1\}) \setminus \bigcap_{n \geq 1} [0, \min\{a + \tfrac{1}{n}, 1\}).$$

  The second equality holds because $a < 1$ gives $\bigcap_{n \geq 1} [0, \min\{a + \frac{1}{n}, 1\}) = [0, a]$. Each set $[0, \min\{b, 1\})$ and $[0, \min\{a + \frac{1}{n}, 1\})$ lies in $\mathcal{P}$. Since $\sigma(\mathcal{P})$ is closed under countable intersections and differences, the set lies in $\sigma(\mathcal{P})$.

These cases exhaust all rationals $a < b$. Thus $\mathcal{B}([0, 1)) \subseteq \sigma(\mathcal{P})$.

Now let $\mu, \nu \in \mathcal{P}([0, 1))$ with $F_\mu = F_\nu$. For $0 \leq t < 1$, $\mu([0, t)) = F_\mu(t) = F_\nu(t) = \nu([0, t))$. Also $\mu([0, 1)) = 1 = \nu([0, 1))$. So $\mu = \nu$ on $\mathcal{P}$, and $\mu([0, 1)) = \nu([0, 1))$. Both are finite measures on $\mathcal{B}([0, 1)) = \sigma(\mathcal{P})$. Therefore, according to the coincidence criterion for finite measures, it holds that $\mu = \nu$ on $\mathcal{B}([0, 1))$.

2.c

Remark (Solution). For $F \in \mathcal{D}$ we construct $\mu_F \in \mathcal{P}([0,1))$ with $F_{\mu_F} = F$. Together with (a) and (b) this shows that $\mu \mapsto F_\mu$ is a bijection $\mathcal{P}([0,1)) \to \mathcal{D}$ with inverse $F \mapsto \mu_F$. Here $\mathbb{N} = \{0, 1, 2, \dots\}$ throughout, and all sequences of intervals are indexed by $\mathbb{N}$.

The extension of F. Since $F \in \mathcal{D}$ is nondecreasing with $F(0) = 0$ and $F(1) = 1$, it takes values in $[F(0), F(1)] = [0,1]$. Define $F \coloneqq 0$ on $(-\infty, 0)$ and $F \coloneqq 1$ on $(1, \infty)$, so $F : \mathbb{R} \to [0,1]$. The extension is nondecreasing. Indeed, let $s \le t$. If $s < 0$, then $F(s) = 0 \le F(t)$. If $t > 1$, then $F(s) \le 1 = F(t)$. Otherwise $s, t \in [0,1]$ and $F(s) \le F(t)$ because $F \in \mathcal{D}$. The extension is left-continuous at every $t \in \mathbb{R}$, that is, $\lim_{s \uparrow t} F(s) = F(t)$. For $t < 0$ and $t > 1$, F is constant on a neighborhood of t. At $t = 0$, $F(s) = 0$ for $s < 0$, so the left limit is $0 = F(0)$. For $t \in (0,1]$, the values $F(s)$ with $s \in (0,t)$ are the original values, so the left limit is $F(t)$ by the definition of $\mathcal{D}$. (The extension is needed because the Lemma below uses points to the left of 0.)

The cost. Let $\mathcal{I} \coloneqq \{[a,b) : 0 \le a \le b \le 1\}$ and $\ell : \mathcal{I} \to [0,1]$, $\ell([a,b)) \coloneqq F(b) - F(a)$. This is well defined. A nonempty $[a,b)$ determines $a = \min[a,b)$ and $b = \sup[a,b)$. The empty set is $[a,a)$ for every a, and $\ell([a,a)) = F(a) - F(a) = 0$ does not depend on a, so $\ell(\emptyset) = 0$. The values lie in $[0,1]$ because $a \le b$ gives $F(b) - F(a) \ge 0$, and F takes values in $[0,1]$ gives $F(b) - F(a) \le 1$. This is the only possible choice: if $F = F_\mu$, then $\mu([a,b)) = \mu([0,b)) - \mu([0,a)) = F(b) - F(a)$. Let $\ell^* : 2^{[0,1)} \to [0,1]$ be the outer measure generated by ℓ:

$$\ell^*(T) \coloneqq \inf\Big\{\sum_{n \in \mathbb{N}} \ell(I_n) : I_n \in \mathcal{I},\ T \subseteq \bigcup_n I_n\Big\}.$$

Let $T \subseteq [0,1)$. The sequence $I_0 = [0,1)$, $I_n = \emptyset$ for $n \ge 1$ consists of sets in $\mathcal{I}$ and satisfies $T \subseteq \bigcup_n I_n$. Its sum is $\ell([0,1)) = 1$. Therefore, according to the definition of $\ell^*(T)$ as an infimum over such sequences, it holds that $\ell^*(T) \le 1$.

Lemma (finite covering inequality). If $K \subseteq \mathbb{N}$ is finite, $c \le d$, and $[c,d] \subseteq \bigcup_{n \in K} (\alpha_n, \beta_n)$ with $\alpha_n < \beta_n$, then $F(d) - F(c) \le \sum_{n \in K} (F(\beta_n) - F(\alpha_n))$.

Proof. Since $c \in [c,d]$, choose $n_0 \in K$ with $c \in (\alpha_{n_0}, \beta_{n_0})$. Suppose $n_i \in K$ is chosen with $\beta_{n_i} > c$. If $\beta_{n_i} \le d$, then $\beta_{n_i} \in (c,d] \subseteq [c,d]$. Hence we can choose $n_{i+1} \in K$ with $\beta_{n_i} \in (\alpha_{n_{i+1}}, \beta_{n_{i+1}})$, that is, $\alpha_{n_{i+1}} < \beta_{n_i} < \beta_{n_{i+1}}$. In particular $\beta_{n_{i+1}} > c$. The numbers $\beta_{n_0} < \beta_{n_1} < \dots$ strictly increase. Hence the indices n_i are distinct. Since K is finite, the procedure stops at some k, and it stops only when $\beta_{n_k} > d$. We have $F(d) \le F(\beta_{n_k})$ since $d < \beta_{n_k}$, and $F(\alpha_{n_0}) \le F(c)$ since $\alpha_{n_0} < c$. For $1 \le i \le k$ we have $F(\beta_{n_{i-1}}) \ge F(\alpha_{n_i})$ since $\alpha_{n_i} < \beta_{n_{i-1}}$. Hence

$$F(d) - F(c) \le F(\beta_{n_k}) - F(\alpha_{n_0}) = (F(\beta_{n_0}) - F(\alpha_{n_0})) + \sum_{i=1}^{k} (F(\beta_{n_i}) - F(\beta_{n_{i-1}})) \le \sum_{i=0}^{k} (F(\beta_{n_i}) - F(\alpha_{n_i})).$$

The last sum is at most the sum over all $n \in K$, since the n_i are distinct elements of K and all terms $F(\beta_n) - F(\alpha_n)$ are nonnegative.

$\ell^* = \ell$ on $\mathcal{I}$. Homework 3, Problem 1 states: $\ell^*(I) = \ell(I)$ for every $I \in \mathcal{I}$ if and only if $\ell(I) \le \sum_n \ell(I_n)$ whenever $I, I_n \in \mathcal{I}$ and $I \subseteq \bigcup_{n \in \mathbb{N}} I_n$. So it suffices to show the covering inequality: for all $0 \le a \le b \le 1$ and all $[a_n, b_n) \in \mathcal{I}$ $(n \in \mathbb{N})$ with $[a,b) \subseteq \bigcup_n [a_n, b_n)$,

$$F(b) - F(a) \le \sum_n (F(b_n) - F(a_n)).$$

If $a = b$ the left side is 0 and the right side is nonnegative, so let $a < b$. Fix $\epsilon > 0$. We show that $F(b) - F(a)$ is at most the right side plus 2ϵ. This suffices because $\epsilon > 0$ is arbitrary. Since $b > a \ge 0$, F is left-continuous at b by the extension step. Hence we can choose $b' \in [a,b)$ with $F(b) - F(b') < \epsilon$. F is left-continuous at each a_n by the extension step. Hence we can choose $a_n' < a_n$ with $F(a_n) - F(a_n') < \epsilon 2^{-n-1}$. Then $\sum_{n \in \mathbb{N}} \epsilon 2^{-n-1} = \epsilon$. (These choices depend on ϵ, F, b and the a_n, and do not depend on the finite subcover chosen below.) Let $x \in [a, b']$. Then $x \in [a,b)$, so $x \in [a_n, b_n)$ for some n, and therefore $a_n' < a_n \le x < b_n$. Thus $[a, b'] \subseteq \bigcup_n (a_n', b_n)$, where each (a_n', b_n) is an open interval with $a_n' < b_n$. The

set $[a, b']$ is compact. Hence a finite set $K \subseteq \mathbb{N}$ satisfies $[a, b'] \subseteq \bigcup_{n\in K}(a'_n, b_n)$. The Lemma with $c = a$, $d = b'$, $\alpha_n = a'_n$, $\beta_n = b_n$ gives $F(b') - F(a) \leq \sum_{n\in K}(F(b_n) - F(a'_n))$. Hence

$$F(b) - F(a) < \epsilon + F(b') - F(a) \leq \epsilon + \sum_{n\in K}(F(b_n) - F(a'_n)) \leq \epsilon + \sum_{n\in\mathbb{N}}(F(b_n) - F(a_n)) + \sum_{n\in\mathbb{N}} \epsilon 2^{-n-1}.$$

The last step uses $F(b_n) - F(a'_n) = (F(b_n) - F(a_n)) + (F(a_n) - F(a'_n))$, then $F(a_n) - F(a'_n) < \epsilon 2^{-n-1}$, and then enlarges K to $\mathbb{N}$. Enlarging K is allowed because $F(b_n) - F(a_n) \geq 0$ and $\epsilon 2^{-n-1} > 0$ for every n. Since $\sum_{n\in\mathbb{N}} \epsilon 2^{-n-1} = \epsilon$, we obtain

$$F(b) - F(a) \leq \sum_{n\in\mathbb{N}}(F(b_n) - F(a_n)) + 2\epsilon.$$

Measurability. By Carathéodory's theorem the ℓ^*-measurable sets form a σ-algebra $\mathcal{M}$ on $[0, 1)$, and $\ell^*|_{\mathcal{M}}$ is a measure. A set $S \subseteq [0, 1)$ lies in $\mathcal{M}$ if and only if $\ell^*(T) = \ell^*(T\cap S)+\ell^*(T\setminus S)$ for every $T \subseteq [0, 1)$. We show $\mathcal{P} \subseteq \mathcal{M}$. Then $\mathcal{I} \subseteq \mathcal{M}$ as well, since $[a, b) = [0, b)\setminus[0, a)$ for $0 \leq a \leq b \leq 1$ and $\mathcal{M}$ is closed under differences. Fix $0 \leq t \leq 1$ and $T \subseteq [0, 1)$. Subadditivity gives $\ell^*(T) \leq \ell^*(T \cap [0, t)) + \ell^*(T \setminus [0, t))$, since $T = (T\cap[0, t))\cup(T\setminus[0, t))$. For the reverse inequality, fix $\epsilon > 0$. We show that $\ell^*(T\cap[0, t))+\ell^*(T\setminus[0, t)) \leq \ell^*(T) + \epsilon$. This suffices because $\epsilon > 0$ is arbitrary. Since $\ell^*(T) \leq 1 < \infty$, the definition of $\ell^*(T)$ as an infimum gives $I_n = [a_n, b_n) \in \mathcal{I}$, $n \in \mathbb{N}$, with $T \subseteq \bigcup_n I_n$ and $\sum_n \ell(I_n) \leq \ell^*(T) + \epsilon$. The sets

$$I_n \cap [0, t) = [a_n, \max\{a_n, \min\{b_n, t\}\}), \qquad I_n \setminus [0, t) = [\min\{b_n, \max\{a_n, t\}\}, b_n)$$

lie in $\mathcal{I}$. Indeed, their endpoints lie between a_n and b_n, hence in $[0, 1]$, and the left endpoint is at most the right endpoint. Since $T \subseteq \bigcup_n I_n$, the first sets cover $T \cap [0, t)$ and the second sets cover $T \setminus [0, t)$. Therefore, according to the definition of ℓ^* as an infimum over covers by sets in $\mathcal{I}$,

$$\ell^*(T \cap [0, t)) \leq \sum_n \ell(I_n \cap [0, t)), \qquad \ell^*(T \setminus [0, t)) \leq \sum_n \ell(I_n \setminus [0, t)).$$

For each n, $\ell(I_n\cap[0, t))+\ell(I_n\setminus[0, t)) = \ell(I_n)$. If $a_n < t < b_n$, the sum is $(F(t)-F(a_n))+(F(b_n)-F(t)) = F(b_n) - F(a_n)$. If $t \leq a_n$, the first set is empty with cost 0 and the second is I_n. If $t \geq b_n$, the first set is I_n and the second is empty with cost 0. Hence

$$\ell^*(T \cap [0, t)) + \ell^*(T \setminus [0, t)) \leq \sum_n \ell(I_n) \leq \ell^*(T) + \epsilon.$$

Conclusion. By the Lemma of (b), $\mathcal{B}([0, 1)) = \sigma(\mathcal{P}) \subseteq \mathcal{M}$, because $\mathcal{M}$ is a σ-algebra containing $\mathcal{P}$. Hence $\mu_F \coloneqq \ell^*|_{\mathcal{B}([0,1))}$ is a measure on $\mathcal{B}([0, 1))$, as a restriction of the measure $\ell^*|_{\mathcal{M}}$ to a sub-σ-algebra. Since $\ell^* = \ell$ on $\mathcal{I}$ and $[0, t) \in \mathcal{I}$, $\mu_F([0, t)) = \ell([0, t)) = F(t) - F(0) = F(t)$ for $0 \leq t < 1$ and $\mu_F([0, 1)) = F(1) - F(0) = 1$. So $\mu_F \in \mathcal{P}([0, 1))$. Also $F_{\mu_F}(t) = \mu_F([0, t)) = F(t)$ for $0 \leq t < 1$ and $F_{\mu_F}(1) = 1 = F(1)$ by definition. Thus $F_{\mu_F} = F$.
Bijection. The map $\mu \mapsto F_\mu$ maps $\mathcal{P}([0, 1))$ into $\mathcal{D}$ by (a). It is injective by (b). It is surjective: for $F \in \mathcal{D}$ we showed $\mu_F \in \mathcal{P}([0, 1))$ and $F_{\mu_F} = F$. Hence it is a bijection. Its inverse is $F \mapsto \mu_F$, since $F_{\mu_F} = F$ for all $F \in \mathcal{D}$ and the map is injective.

Remark (Discussion).
- Passing from a countable to a finite cover requires compactness, which needs closed bounded sets. The cover of $[a, b)$ by half-open sets is converted to a cover of $[a, b']$ by the open sets (a'_n, b_n). This needs two modifications. First, replace b by $b' < b$ (shrinking the target at its right end). Second, replace each a_n by $a'_n < a_n$ (enlarging each covering set at its left end). Left-continuity of F is exactly what makes both modifications cost an arbitrarily small amount. This is why left-continuous F pairs with intervals $[a, b)$.

- The jump $\lim_{s\downarrow t} F(s) - F(t)$ equals $\mu(\{t\})$. Hence F need not be right-continuous, and the atoms of μ are the discontinuities of F.

3. (Borel sets for a metric outer measure)

3.a

Remark (Minimal solution). Let (H) denote the hypothesis: $\mu^*(A\cup B) = \mu^*(A)+\mu^*(B)$ if $\operatorname{dist}(A,B) > 0$. Closed sets generate $\mathcal{B}(X)$ and the μ^*-measurable sets form a σ-algebra (Carathéodory). Hence it suffices to show that every closed $C \neq \emptyset$ is measurable, since $\emptyset$ is measurable. By subadditivity, $\mu^*(E) \leq \mu^*(E\cap C)+\mu^*(E\setminus C)$. Hence it suffices to show $\mu^*(E\cap C)+\mu^*(E\setminus C) \leq \mu^*(E)$ for every $E \subseteq X$. This holds if $\mu^*(E) = \infty$, so let $\mu^*(E) < \infty$. Let $E_k \coloneqq \{x \in E : \operatorname{dist}(x,C) > 2^{-k}\}$ and $D_j \coloneqq \{x \in E \setminus C : 2^{-j-1} < \operatorname{dist}(x,C) \leq 2^{-j}\}$. For $x \in E \cap C$ and $y \in E_k$, $d(x,y) \geq \operatorname{dist}(y,C) > 2^{-k}$, so $\operatorname{dist}(E\cap C, E_k) \geq 2^{-k} > 0$. Hence (H) and monotonicity give $\mu^*(E\cap C)+\mu^*(E_k) = \mu^*((E\cap C)\cup E_k) \leq \mu^*(E)$. Let $x \in E \setminus C$. Since C is closed, $\operatorname{dist}(x,C) \in (0,\infty)$. If $\operatorname{dist}(x,C) > 2^{-k}$, then $x \in E_k$. If $\operatorname{dist}(x,C) \leq 2^{-k}$, then $x \in D_j$ for the j with $2^{-j-1} < \operatorname{dist}(x,C) \leq 2^{-j}$, and $2^{-j-1} < 2^{-k}$ gives $j \geq k$. Hence $E \setminus C = E_k \cup \bigcup_{j\geq k} D_j$. By countable subadditivity, $\mu^*(E\setminus C) \leq \mu^*(E_k) + \sum_{j\geq k}\mu^*(D_j)$. As $\operatorname{dist}(\cdot,C)$ is 1-Lipschitz, for $j \geq i+2$, $x \in D_i$, $y \in D_j$ we get $d(x,y) \geq \operatorname{dist}(x,C) - \operatorname{dist}(y,C) > 2^{-i-1} - 2^{-j} \geq 2^{-i-1} - 2^{-i-2} = 2^{-i-2}$. Let $r \in \{0,1\}$ and $m \geq 1$. For $i < m$ we have $(2m+r)-(2i+r) \geq 2$ and $2i+r \leq 2m-1$, so $d(x,y) \geq 2^{-(2i+r)-2} \geq 2^{-2m-1}$ for $x \in D_{2i+r}$, $y \in D_{2m+r}$. Hence $\operatorname{dist}(D_{2m+r}, \bigcup_{i<m} D_{2i+r}) \geq 2^{-2m-1} > 0$. By (H), $\mu^*(\bigcup_{i\leq m} D_{2i+r}) = \mu^*(D_{2m+r}) + \mu^*(\bigcup_{i<m} D_{2i+r})$. By induction on m and monotonicity, $\sum_{i=0}^{m}\mu^*(D_{2i+r}) = \mu^*(\bigcup_{i\leq m} D_{2i+r}) \leq \mu^*(E)$ for all $m \in \mathbb{N}$. Letting $m \to \infty$ gives $\sum_{i\in\mathbb{N}}\mu^*(D_{2i+r}) \leq \mu^*(E)$ for $r = 0, 1$. Adding the cases gives $\sum_j \mu^*(D_j) \leq 2\mu^*(E) < \infty$. Therefore, for every k,

$$\mu^*(E\cap C) + \mu^*(E\setminus C) \leq \mu^*(E\cap C) + \mu^*(E_k) + \sum_{j\geq k}\mu^*(D_j) \leq \mu^*(E) + \sum_{j\geq k}\mu^*(D_j).$$

The tails $\sum_{j\geq k}\mu^*(D_j)$ of the convergent series $\sum_j \mu^*(D_j)$ tend to 0 as $k \to \infty$. The left side does not depend on k. Letting $k \to \infty$ gives $\mu^*(E\cap C) + \mu^*(E\setminus C) \leq \mu^*(E)$.

Remark (Solution). Hypothesis (H): $\mu^*(A\cup B) = \mu^*(A)+\mu^*(B)$ whenever $A, B \subseteq X$ and $\operatorname{dist}(A,B) > 0$. The μ^*-measurable sets form a σ-algebra by Carathéodory's theorem. The σ-algebra $\mathcal{B}(X)$ is generated by the closed sets. Hence it suffices to show that every closed $C \subseteq X$ is μ^*-measurable, that is, $\mu^*(E) = \mu^*(E\cap C) + \mu^*(E\setminus C)$ for every $E \subseteq X$. The empty set is measurable, so let $C \neq \emptyset$. Then $\operatorname{dist}(x,C) < \infty$ for all $x \in X$. Fix a test set $E \subseteq X$. Subadditivity gives $\mu^*(E) \leq \mu^*(E\cap C) + \mu^*(E\setminus C)$. It remains to show the reverse inequality $\mu^*(E\cap C) + \mu^*(E\setminus C) \leq \mu^*(E)$. If $\mu^*(E) = \infty$ it holds because the right side is infinite, so assume $\mu^*(E) < \infty$. For $k, j \in \mathbb{N}$ let

$$E_k \coloneqq \{x \in E : \operatorname{dist}(x,C) > 2^{-k}\}, \qquad D_j \coloneqq \{x \in E\setminus C : 2^{-j-1} < \operatorname{dist}(x,C) \leq 2^{-j}\}.$$

Step 1: $\mu^*(E\cap C) + \mu^*(E_k) \leq \mu^*(E)$ for every k. For $x \in C$ and $y \in E_k$, $d(x,y) \geq \operatorname{dist}(y,C) > 2^{-k}$, so $\operatorname{dist}(E\cap C, E_k) \geq 2^{-k} > 0$. By (H), $\mu^*(E\cap C)+\mu^*(E_k) = \mu^*((E\cap C)\cup E_k)$. Since $(E\cap C)\cup E_k \subseteq E$, monotonicity gives $\mu^*((E\cap C)\cup E_k) \leq \mu^*(E)$.
Step 2: $\mu^*(E\setminus C) \leq \mu^*(E_k) + \sum_{j\geq k}\mu^*(D_j)$ for every k. Let $x \in E\setminus C$. Since C is closed and $x \notin C$, $\operatorname{dist}(x,C) > 0$. Hence x falls into one of two cases. If $\operatorname{dist}(x,C) > 2^{-k}$, then $x \in E_k$. If $0 < \operatorname{dist}(x,C) \leq 2^{-k}$, then there is j with $2^{-j-1} < \operatorname{dist}(x,C) \leq 2^{-j}$, so $x \in D_j$. Moreover $2^{-j-1} < 2^{-k}$, so $j \geq k$. So $E\setminus C = E_k \cup \bigcup_{j\geq k} D_j$. Countable subadditivity gives $\mu^*(E\setminus C) \leq \mu^*(E_k) + \sum_{j\geq k}\mu^*(D_j)$.
Step 3: $\operatorname{dist}(D_i, D_j) \geq 2^{-i-2}$ whenever $j \geq i+2$. Let $x \in D_i$, $y \in D_j$. For every $z \in C$, $\operatorname{dist}(x,C) \leq$

$d(x,z) \le d(x,y) + d(y,z)$. Taking the infimum over $z \in C$ gives $\operatorname{dist}(x,C) \le d(x,y) + \operatorname{dist}(y,C)$. Hence

$$d(x,y) \ge \operatorname{dist}(x,C) - \operatorname{dist}(y,C) > 2^{-i-1} - 2^{-j} \ge 2^{-i-1} - 2^{-i-2} = 2^{-i-2},$$

where $j \ge i+2$ was used in the last inequality. Therefore $\operatorname{dist}(D_i, D_j) \ge 2^{-i-2} > 0$.
Step 4: $\sum_j \mu^*(D_j) \le 2\mu^*(E) < \infty$**.** Let $r \in \{0,1\}$ and $m \ge 1$. For $i < m$ we have $(2m+r)-(2i+r) \ge 2$ and $2i + r \le 2m - 1$. By Step 3, $\operatorname{dist}(D_{2i+r}, D_{2m+r}) \ge 2^{-(2i+r)-2} \ge 2^{-2m-1}$. Let $x \in D_{2m+r}$ and $y \in \bigcup_{i<m} D_{2i+r}$. Then $y \in D_{2i+r}$ for some $i < m$, so $d(x,y) \ge 2^{-2m-1}$. Hence $\operatorname{dist}(D_{2m+r}, \bigcup_{i<m} D_{2i+r}) \ge 2^{-2m-1} > 0$. By (H), $\mu^*(\bigcup_{i\le m} D_{2i+r}) = \mu^*(D_{2m+r}) + \mu^*(\bigcup_{i<m} D_{2i+r})$. By induction on m (the case $m = 0$ is $\mu^*(D_r) = \mu^*(D_r)$) and monotonicity, $\sum_{i=0}^{m} \mu^*(D_{2i+r}) = \mu^*(\bigcup_{i\le m} D_{2i+r}) \le \mu^*(E)$ for all $m \in \mathbb{N}$, since the union is contained in E. Letting $m \to \infty$ gives $\sum_{i\in\mathbb{N}} \mu^*(D_{2i+r}) \le \mu^*(E)$ for $r = 0, 1$. Adding the cases $r = 0, 1$ gives $\sum_{j\in\mathbb{N}} \mu^*(D_j) \le 2\mu^*(E) < \infty$.
Step 5: conclusion. By Step 2 and then Step 1, for every k,

$$\mu^*(E\cap C) + \mu^*(E\setminus C) \le \mu^*(E\cap C) + \mu^*(E_k) + \sum_{j\ge k} \mu^*(D_j) \le \mu^*(E) + \sum_{j\ge k} \mu^*(D_j).$$

The left side and $\mu^*(E)$ do not depend on k. By Step 4, the tail $\sum_{j\ge k} \mu^*(D_j)$ of a convergent series tends to 0 as $k \to \infty$. Hence $\mu^*(E\cap C) + \mu^*(E\setminus C) \le \mu^*(E)$.

Remark (Discussion).

- The hypothesis only gives additivity for sets at positive distance, but $E \setminus C$ touches C. The dyadic annuli D_j cut $E \setminus C$ into pieces that approach C. Step 4 uses only the finiteness $\mu^*(E) < \infty$ to show that their total outer measure $\sum_j \mu^*(D_j)$ is finite. Hence the pieces nearest to C have total outer measure $\sum_{j\ge k} \mu^*(D_j) \to 0$ as $k \to \infty$.
- Neighbouring annuli are not separated. This is why even and odd indices are treated separately. The annuli D_i, D_j with $|i-j| \ge 2$ are separated by a gap depending on the smaller index.

4. (Uniqueness under σ-finiteness)

4.a

Remark (Minimal solution). Index the sets $P_n \in \mathcal{P}$ by $n \in \mathbb{N}$, so that $\bigcup_n P_n = X$ and $\mu(P_n) = \nu(P_n) < \infty$. For each n let $\mu_n(A) \coloneqq \mu(A \cap P_n)$ and $\nu_n(A) \coloneqq \nu(A \cap P_n)$ for $A \in \mathcal{E}$. These are measures on $\mathcal{E} = \sigma(\mathcal{P})$ with $\mu_n(X) = \mu(P_n) = \nu(P_n) = \nu_n(X) < \infty$. For $P \in \mathcal{P}$, $P \cap P_n \in \mathcal{P}$ since $\mathcal{P}$ is a π-system, so $\mu_n(P) = \mu(P \cap P_n) = \nu(P \cap P_n) = \nu_n(P)$. By the coincidence criterion for finite measures, $\mu_n = \nu_n$ on $\mathcal{E}$. Let $Q_n \coloneqq P_n \setminus \bigcup_{m<n} P_m \in \mathcal{E}$. These are pairwise disjoint, $Q_n \subseteq P_n$, and $\bigcup_n Q_n = \bigcup_n P_n = X$. For $A \in \mathcal{E}$ and $n \in \mathbb{N}$, $(A \cap Q_n) \cap P_n = A \cap Q_n$, so $\mu(A \cap Q_n) = \mu_n(A \cap Q_n)$ and $\nu(A \cap Q_n) = \nu_n(A \cap Q_n)$. By countable additivity, for $A \in \mathcal{E}$, $\mu(A) = \sum_n \mu(A \cap Q_n) = \sum_n \mu_n(A \cap Q_n) = \sum_n \nu_n(A \cap Q_n) = \sum_n \nu(A \cap Q_n) = \nu(A)$.

Remark (Solution). We show that $\mu(A) = \nu(A)$ for every $A \in \mathcal{E}$. The hypotheses are: $\mathcal{P}$ is a π-system with $\sigma(\mathcal{P}) = \mathcal{E}$, $\mu = \nu$ on $\mathcal{P}$, and $P_n \in \mathcal{P}$ satisfy $\bigcup_n P_n = X$ and $\mu(P_n) = \nu(P_n) < \infty$. Index the sets P_n by $n \in \mathbb{N}$. Step 1 shows $\mu(A \cap P_n) = \nu(A \cap P_n)$ for all $n \in \mathbb{N}$ and $A \in \mathcal{E}$, using the coincidence criterion for finite measures. Step 2 deduces $\mu = \nu$ by countable additivity over a disjoint decomposition of X.
Step 1: finite pieces. Fix n and set $\mu_n(A) \coloneqq \mu(A \cap P_n)$, $\nu_n(A) \coloneqq \nu(A \cap P_n)$ for $A \in \mathcal{E}$. These are well defined since $P_n \in \mathcal{P} \subseteq \mathcal{E}$, so $A \cap P_n \in \mathcal{E}$. They are measures on $\mathcal{E} = \sigma(\mathcal{P})$: $\mu_n(\emptyset) = 0$, and if $A_i \in \mathcal{E}$ are pairwise disjoint, then the sets $A_i \cap P_n$ are pairwise disjoint and $\mu_n(\bigcup_i A_i) = \sum_i \mu(A_i \cap P_n) = \sum_i \mu_n(A_i)$.

The same holds for ν_n. Moreover $\mu_n(X) = \mu(P_n) = \nu(P_n) = \nu_n(X) < \infty$, so μ_n, ν_n are finite. For $P \in \mathcal{P}$ we have $P \cap P_n \in \mathcal{P}$ since $\mathcal{P}$ is a π-system, so $\mu_n(P) = \mu(P \cap P_n) = \nu(P \cap P_n) = \nu_n(P)$. Therefore μ_n, ν_n are finite measures on $\sigma(\mathcal{P})$ that agree on $\mathcal{P}$ and satisfy $\mu_n(X) = \nu_n(X)$. According to the coincidence criterion for finite measures, it holds that $\mu_n = \nu_n$ on $\mathcal{E}$.

Step 2: summation. Let $Q_n \coloneqq P_n \setminus \bigcup_{m<n} P_m \in \mathcal{E}$. The Q_n are pairwise disjoint, $Q_n \subseteq P_n$, and $\bigcup_n Q_n = \bigcup_n P_n = X$. Let $A \in \mathcal{E}$. Since $A \cap Q_n \subseteq P_n$, we have $(A \cap Q_n) \cap P_n = A \cap Q_n$, hence $\mu(A \cap Q_n) = \mu_n(A \cap Q_n)$ and $\nu(A \cap Q_n) = \nu_n(A \cap Q_n)$. The sets $A \cap Q_n \in \mathcal{E}$ are pairwise disjoint with union A. Hence, by countable additivity and Step 1,

$$\mu(A) = \sum_n \mu(A \cap Q_n) = \sum_n \mu_n(A \cap Q_n) = \sum_n \nu_n(A \cap Q_n) = \sum_n \nu(A \cap Q_n) = \nu(A).$$